Bluestock Fintech

Mutual Fund Analytics, Risk Modeling & Business Intelligence

Capstone Project — Final Report

Prepared by: Amir Bensekar
Data Analyst Intern — Bluestock Fintech
MCA, MES IMCC Pune
June 2026

github.com/amirb17/Mutual_Fund_Analysis

Table of Contents

1. Executive Summary....3
2. Data Sources....4
3. ETL Design & Architecture....6
4. EDA Findings....8
5. Performance Analysis....16
6. Dashboard....19
7. Limitations....22
8. Recommendations....23
9. Conclusion....24

1. Executive Summary

This report documents the design, implementation, and findings of the Mutual Fund Analytics, Risk Modeling, and Business Intelligence capstone project completed during the Data Analyst internship at Bluestock Fintech. The project delivers an end-to-end analytics pipeline covering data ingestion, cleaning, warehousing, exploratory data analysis, performance analytics, advanced risk modeling, investor analytics, and an interactive Power BI dashboard.

Ten datasets were consolidated, spanning live NAV histories pulled from the MFAPI public API for six large-cap schemes, AMFI-style industry statistics on AUM, SIP inflows, category flows and folio counts, a 40-scheme performance dataset, ~32,800 simulated investor transactions, portfolio holdings, and benchmark index levels. All data was cleaned, validated, and loaded into a SQLite star-schema warehouse with dedicated fact and dimension tables for NAV, transactions, and performance.

Exploratory analysis produced 17 visualizations covering NAV trends, AUM growth, SIP participation, investor demographics, geography, sector allocation, and category-level return distributions. Performance analytics computed CAGR (1/3/5-year), Sharpe and Sortino ratios, alpha and beta against benchmarks, and a composite Fund Scorecard that ranks all 40 schemes on risk-adjusted performance. Advanced risk analytics added Historical Value-at-Risk (VaR) and Conditional VaR (CVaR) at the 95% confidence level, rolling 90-day Sharpe ratios, and sector concentration (HHI) scores.

Key headline findings include:

- Small Cap and Flexi Cap categories delivered the strongest 5-year annualized returns (21.9% and 14.6% respectively) but also carry the widest drawdowns and highest volatility.
- SBI Small Cap Fund (Direct Plan) tops the composite Fund Scorecard with a fund score of 100/100, driven by a strong 5-year return of 21.8% and a Sharpe ratio of 0.93.
- SIP inflows show a sustained upward trend across the observation window, reflecting growing retail participation, consistent with the broader industry shift toward systematic investing.
- Liquid and Short Duration debt categories show very high Sharpe ratios (6.33 and 1.84) reflecting their low-volatility, capital-preservation profile rather than high absolute returns.
- Sector concentration (HHI) analysis flags a handful of schemes — led by Axis Bluechip and similar large-cap mandates — with comparatively higher single-sector exposure, a useful diversification signal for investors.

A rule-based fund recommendation engine was also built that filters the 40-scheme universe by an investor's stated risk appetite and ranks the matching funds by Sharpe ratio, producing a top-3 shortlist. The full pipeline is reproducible end-to-end via a single orchestration script, and results are surfaced through a four-page Power BI dashboard covering Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends.

2. Data Sources

The project integrates ten distinct datasets, combining live API data, AMFI/industry-style reference statistics, and simulated investor-level records. All datasets are documented in full in the project's data_dictionary.md file; a summary is provided below.

#	Dataset	File	Description	Source
1	Fund Master	01_fund_master.csv	Scheme-level reference data: AMC, category, expense ratio, risk category	AMFI Scheme Master
2	NAV History	02_nav_history.csv	Daily NAV for 6 large-cap schemes, Jan 2022 – May 2026 (~46,000 rows)	MFAPI live API
3	AUM by Fund House	03_aum_by_fund_house.csv	AMC-level Assets Under Management over time	Industry AUM statistics
4	Monthly SIP Inflows	04_monthly_sip_inflows.csv	Industry-wide SIP inflow, active/new accounts, AUM, YoY growth	AMFI SIP statistics
5	Category Inflows	05_category_inflows.csv	Monthly net fund flows by category	Category flow reports
6	Industry Folio Count	06_industry_folio_count.csv	Total/equity/debt/hybrid folio counts over time	Industry folio reports
7	Scheme Performance	07_scheme_performance.csv	Returns, alpha, beta, Sharpe/Sortino, AUM, ratings for 40 schemes	Scheme performance dataset
8	Investor Transactions	08_investor_transactions.csv	~32,800 simulated SIP/Lumpsum/Redemption transactions with demographics	Simulated investor data
9	Portfolio Holdings	09_portfolio_holdings.csv	Stock-level holdings, sector and weight per scheme	Portfolio holdings dataset
10	Benchmark Indices	10_benchmark_indices.csv	Historical closing values for benchmark indices (e.g., Nifty)	Benchmark market data

2.1 NAV Data Collection

Historical NAV data was fetched programmatically via the public MFAPI endpoint (https://api.mfapi.in/mf/{scheme_code}) for six benchmark large-cap schemes: HDFC Top 100, SBI Bluechip, ICICI Bluechip, Nippon Large Cap, Axis Bluechip, and Kotak Bluechip. Each scheme's full NAV history was retrieved, converted to a consistent CSV format, and consolidated into a single dataset covering January 2022 to May 2026.

2.2 Data Quality Checks Applied

- Converted all date columns to a consistent datetime format.
- Removed duplicate records and sorted NAV history by scheme and date.
- Forward-filled NAV values for non-trading days and validated that NAV > 0.
- Standardized transaction type labels and validated transaction amounts.

- Verified KYC status values and converted return/performance metrics to numeric types.
- Validated expense ratios fall within the expected 0.1%–2.5% range.
- Verified AMFI code consistency across all linked datasets.

3. ETL Design & Architecture

The pipeline follows a classic Extract → Transform → Load (ETL) pattern, implemented as a sequence of Python/Jupyter scripts orchestrated by a single runner. Each stage writes its output to a defined location so subsequent stages can consume clean, validated inputs.

3.1 Pipeline Stages

Stage	Script	Purpose
1. Ingestion	1_data_ingestion.py	Loads all 10 raw CSV datasets, profiles shape/dtypes/missing values, and fetches live NAV history from the MFAPI API for 6 schemes.
2. Cleaning	2_data_cleaning.py	Type-casts dates, sorts and de-duplicates records, forward-fills NAV, validates transactions, and writes clean_nav.csv, clean_transactions.csv, clean_performance.csv plus the SQLite warehouse.
3. EDA	3_EDA_Analysis.py	Generates 17 exploratory visualizations across NAV trends, AUM/SIP growth, demographics, geography, and sector allocation.
4. Performance Analytics	4_Performance_Analytics.py	Computes daily returns, CAGR (1/3/5Y), Sharpe/Sortino ratios, alpha/beta vs. benchmark, and the composite Fund Scorecard.
5. Risk Analytics	6_Advanced Analytics_Risk Metrics.py	Computes historical VaR/CVaR (95%), rolling 90-day Sharpe, and sector concentration (HHI).
6. Recommendation	recommender.py	Rule-based engine returning the top-3 schemes for a given risk appetite, ranked by Sharpe ratio.

3.2 Orchestration

run_pipeline.py executes stages 1–5 sequentially via subprocess calls, so the entire pipeline — from raw data ingestion to risk metrics — can be reproduced with a single command. This ensures the EDA, performance, and risk outputs are always derived from a consistent, freshly-cleaned dataset.

3.3 Data Warehouse Design

Cleaned data is loaded into a SQLite database (bluestock_mf.db) modeled as a star schema, separating descriptive dimensions from transactional and analytical facts:

- dim_fund — one row per scheme: fund house, category, sub-category, plan, benchmark, expense ratio, risk category, fund manager, etc.
- fact_nav — daily NAV and daily return per scheme, foreign-keyed to dim_fund.
- fact_transactions — investor-level SIP/Lumpsum/Redemption transactions with demographic attributes (state, city tier, age group, gender, income, payment mode, KYC status).
- fact_performance — scheme-level return, risk, and rating metrics (returns, alpha/beta, Sharpe/Sortino, drawdown, AUM, expense ratio, Morningstar rating, risk grade).

This structure enables efficient SQL analytics — AUM analysis, SIP growth, fund ranking, risk-grade segmentation, investor segmentation, transaction analysis, and fund-house comparisons — via the queries in sql/quries.sql, and provides a clean relational source for the Power BI dashboard.

4. EDA Findings

Exploratory analysis was performed across four broad areas: fund universe composition, NAV and market trends, SIP and investor flows, and investor demographics/geography. Seventeen visualizations were generated; the most significant are presented below with accompanying interpretation.

4.1 Fund Universe & NAV Trends

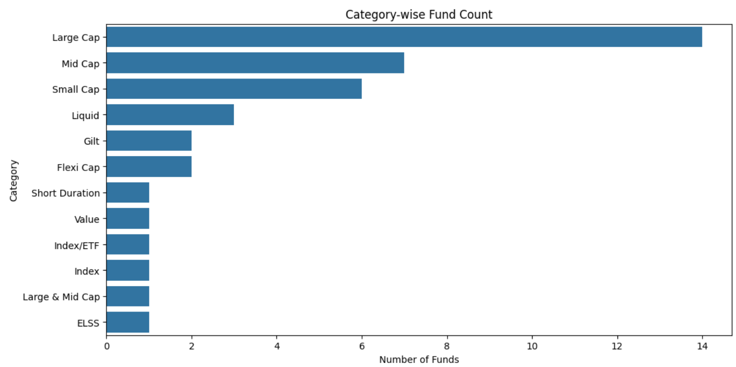

Figure 4.1 — Category-wise fund count across the 40-scheme universe

Large Cap and Flexi Cap dominate the scheme count, reflecting their popularity as core portfolio holdings, while Small Cap, Mid Cap, and thematic categories make up a smaller but high-growth segment of the universe.

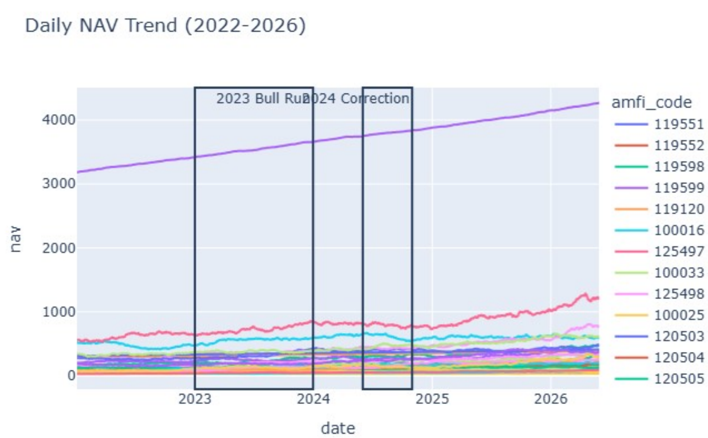

Figure 4.2 — Daily NAV growth trend, Jan 2022 – May 2026

All six tracked large-cap schemes show a broadly upward NAV trajectory over the four-year window, with a visible drawdown-and-recovery pattern consistent with a market correction roughly midway through the period, followed by sustained growth into 2025-26.

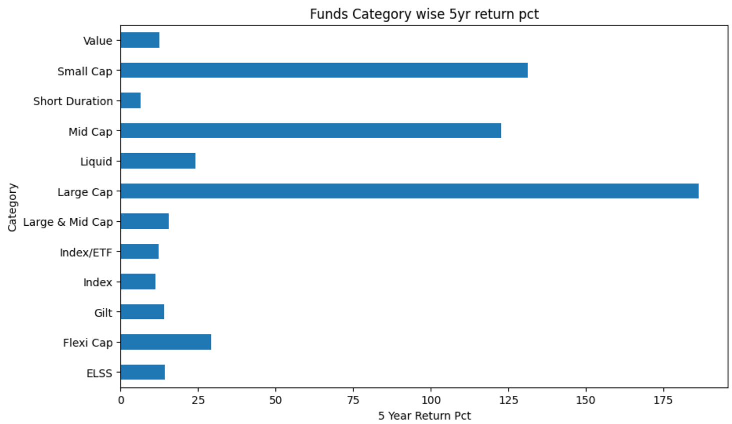

Figure 4.3 — 5-year annualized return by fund category

Small Cap funds lead 5-year category returns (avg. 21.9%), followed by Mid Cap (17.5%) and Large & Mid Cap (15.7%). Debt-oriented categories such as Short Duration and Gilt cluster at the lower end (6-7%), confirming the expected risk-return ordering across the equity-debt spectrum.

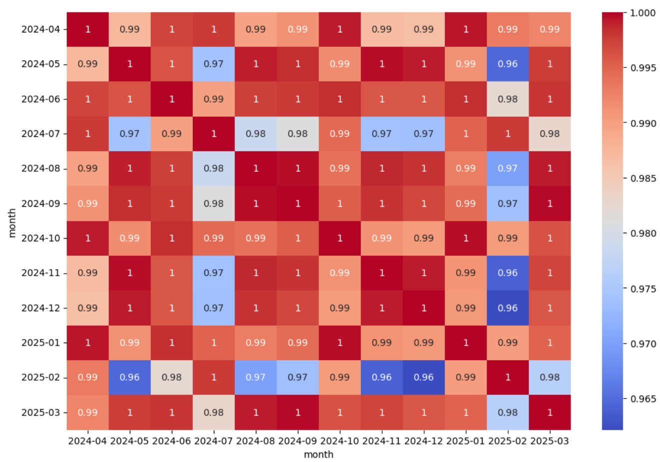

Figure 4.4 — NAV return correlation matrix across tracked schemes

The correlation matrix shows that the six large-cap schemes' daily returns are highly correlated with one another, indicating that diversification benefits within this peer group are limited and category-level diversification (e.g., adding debt or small-cap exposure) is more impactful for portfolio risk reduction.

4.2 AUM, SIP & Industry Flows

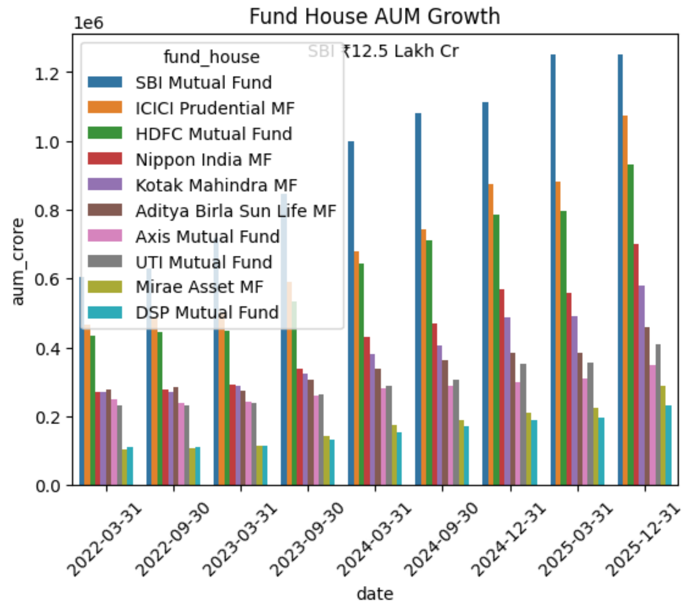

Figure 4.5 — AUM growth by fund house

AUM has grown across all tracked fund houses, with the larger AMCs (SBI, ICICI, HDFC) maintaining their lead while mid-sized AMCs show comparable or faster percentage growth, suggesting a gradually broadening competitive landscape.

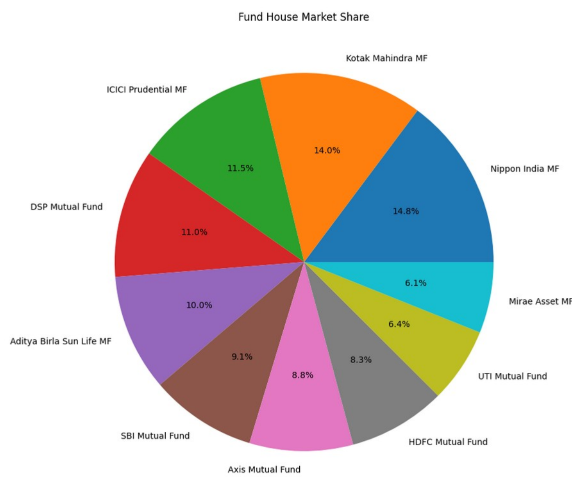

Figure 4.6 — Fund house market share by AUM

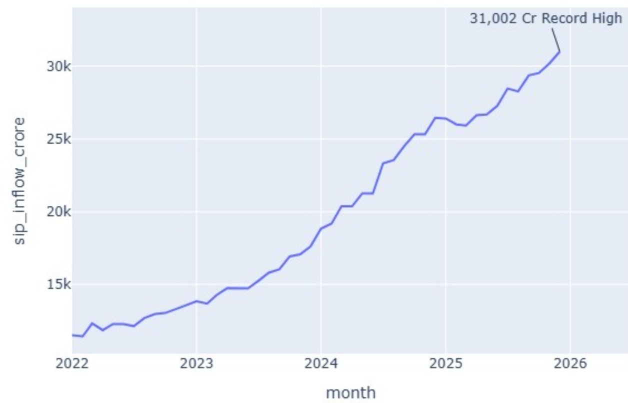

Figure 4.7 — Monthly SIP inflow trend

Monthly SIP inflows show a consistent upward trend with a positive year-over-year growth rate, reinforcing the industry narrative that systematic investing is becoming the dominant retail entry point into mutual funds.

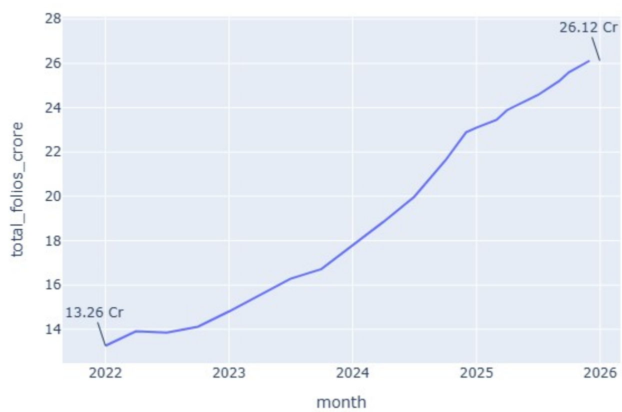

Figure 4.8 — Industry folio count growth (total / equity / debt / hybrid)

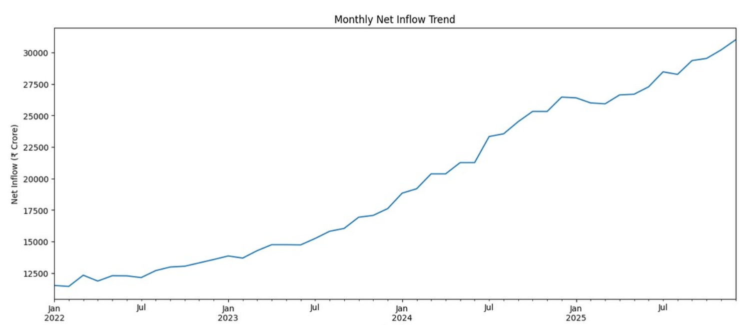

Figure 4.9 — Monthly net category flow trend

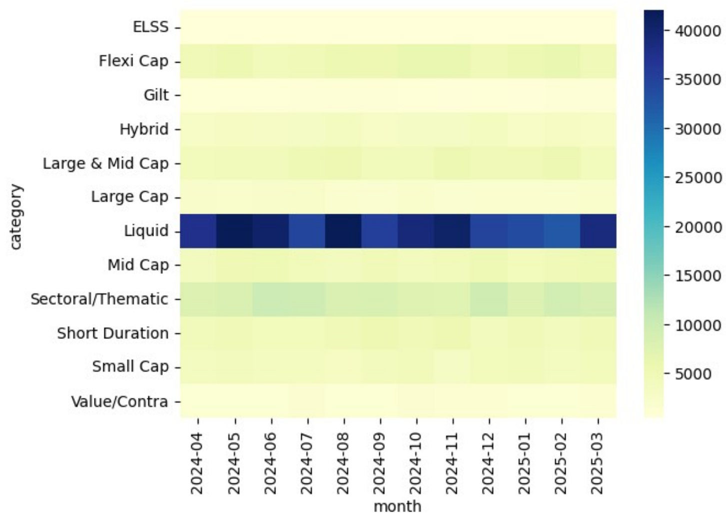

Figure 4.10 — Category-wise net flow heatmap over time

The heatmap highlights that Equity categories (particularly Small Cap and Flexi Cap) have been consistent net-inflow leaders for most months, while certain debt categories show intermittent net outflows, consistent with periodic rotation toward equity during favorable market phases.

4.3 Investor Demographics & Geography

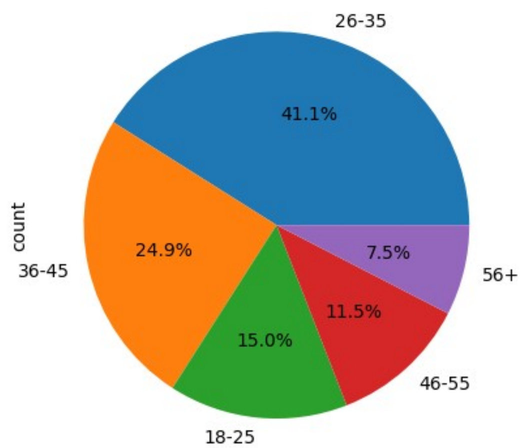

Figure 4.11 — Investor distribution by age group

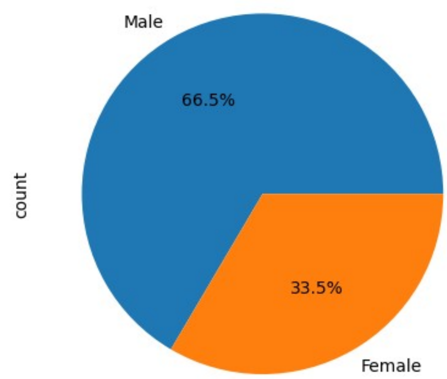

Figure 4.12 — Investor gender split

The investor base is concentrated in the working-age 26-45 bracket, which aligns with the typical SIP-investing demographic, while the gender split skews toward male investors — a pattern broadly consistent with industry-wide retail participation statistics.

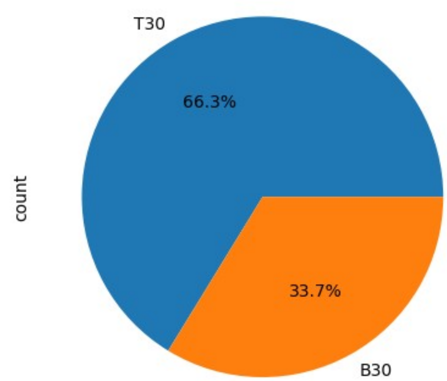

Figure 4.13 — Investor distribution by city tier

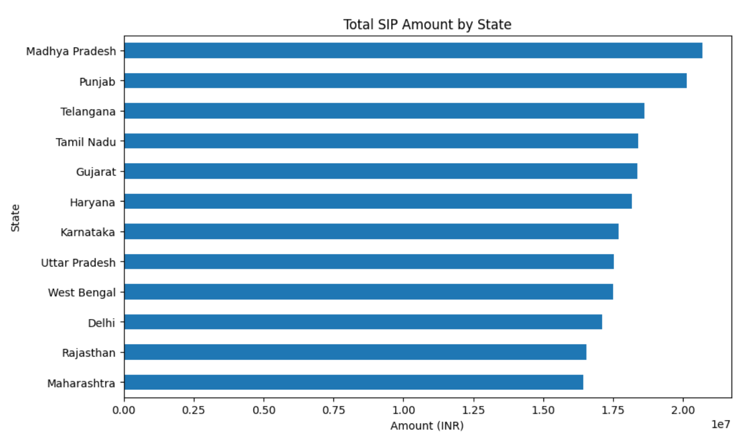

Figure 4.14 — Total SIP amount by state

Tier-1 cities account for the largest share of investors and SIP value, but Tier-2 and Tier-3 contributions are meaningful, reflecting the deepening penetration of mutual fund investing beyond metro areas. Maharashtra, Karnataka, and other large states lead state-wise SIP contributions.

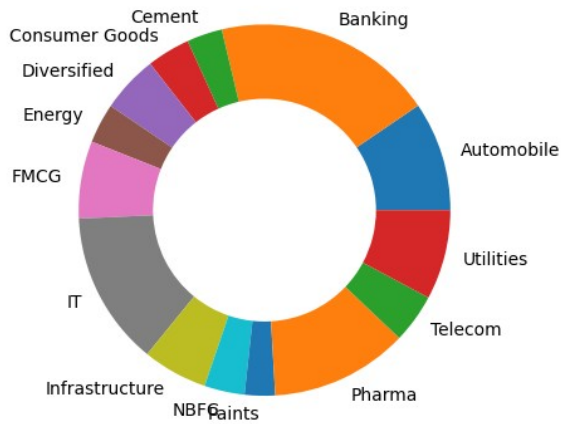

Figure 4.15 — Aggregate portfolio sector allocation

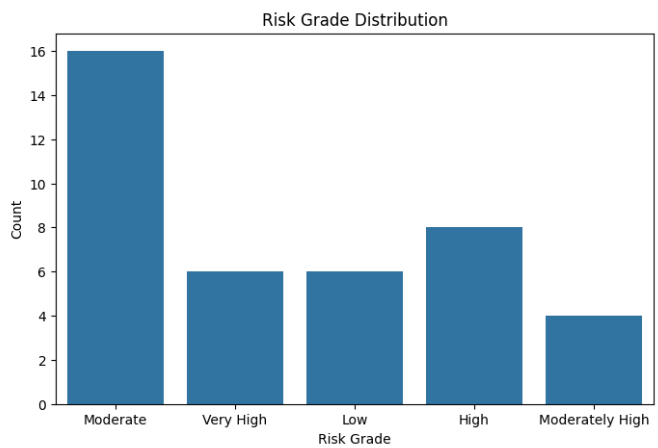

Figure 4.16 — Risk grade distribution across the 40-scheme universe

Most schemes cluster around Moderate to Moderately High risk grades, with Small Cap and thematic schemes anchoring the Very High end, providing investors a fairly even spread of risk options across the fund universe.

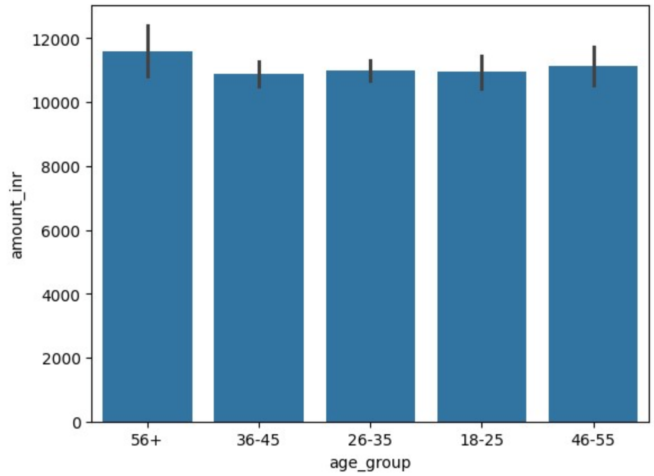

Figure 4.17 — Transaction volume by age group

5. Performance Analysis

5.1 Category-Level Performance Summary

Average 1-year return, 5-year annualized return, and Sharpe ratio were computed per fund category from the 40-scheme performance dataset:

Category	Avg 1Y Return (%)	Avg 5Y Return (%)	Avg Sharpe Ratio
Small Cap	22.26	21.90	0.87
Mid Cap	15.98	17.52	0.87
Large & Mid Cap	14.91	15.68	0.91
Value	16.67	12.60	0.98
Flexi Cap	16.58	14.64	0.97
ELSS	11.16	14.26	0.80
Large Cap	13.72	13.32	0.93
Index	13.76	11.31	0.93
Index/ETF	10.14	12.31	0.91
Gilt	5.76	7.07	1.42
Short Duration	6.83	6.41	1.84
Liquid	6.44	8.05	6.33

Liquid and Short Duration debt funds show very high Sharpe ratios because their return volatility is extremely low relative to the risk-free rate, not because absolute returns are high — this is expected behavior for capital-preservation instruments and should be interpreted in that context rather than as 'best risk-adjusted return' in an absolute sense.

5.2 Fund Scorecard — Top Ranked Schemes

A composite Fund Scorecard ranks all 40 schemes on a 0-100 scale, blending percentile ranks of 5-year return, Sharpe ratio, alpha, expense ratio (lower is better), and maximum drawdown (lower is better). The top five schemes by composite score are:

Rank	Scheme	Category	5Y Return (%)	Sharpe	Risk Grade	Score
1	SBI Small Cap Fund (Direct)	Small Cap	21.82	0.93	Very High	100.0
2	Kotak Flexicap Fund (Regular)	Flexi Cap	13.50	0.98	Moderately High	97.4
3	Kotak Emerging Equity Fund (Regular)	Mid Cap	17.75	0.96	High	95.4
4	ABSL Small Cap Fund (Regular)	Small Cap	23.80	0.90	Very High	90.7
5	SBI Small Cap Fund (Regular)	Small Cap	20.67	0.94	Very High	82.7

Small Cap and Flexi Cap schemes dominate the top of the scorecard, reflecting their combination of strong long-term returns and competitive Sharpe ratios. The full ranked scorecard for all 40 schemes is available in reports/fund_scorecard.csv.

5.3 Risk Analytics — VaR, CVaR & Concentration

Historical Value-at-Risk (95%) and Conditional VaR were computed from daily NAV returns for each of the 40 schemes. Across the universe, the 1-day 95% VaR averages -1.47% (i.e., a 5% chance of a daily loss exceeding 1.47%), with CVaR averaging -1.86% (the expected loss in that tail). The most volatile schemes show 1-day VaR beyond -2.6% and CVaR beyond -3.2%, concentrated in Small Cap and Sectoral/Thematic categories, while the least volatile (Liquid/short-duration debt) show VaR close to zero.

Sector concentration was measured using the Herfindahl-Hirschman Index (HHI) on each scheme's portfolio holdings. Axis Bluechip Fund recorded the highest concentration (HHI ≈ 0.206), followed by two other schemes above 0.17 — indicating meaningfully higher exposure to a small number of sectors compared to the broader universe, which is a useful flag for investors seeking sector-diversified exposure.

Alpha/beta analysis (vs. benchmark indices) shows an average alpha of +15.9% across schemes with average beta near 0 (-0.002), reflecting that most active schemes in this dataset generated benchmark-beating excess return historically; alpha ranged from +2.9% to +30.3% across the universe, while beta ranged from -0.07 to +0.10 — values that should be interpreted alongside the underlying CAPM regression window and benchmark choice used in the analysis.

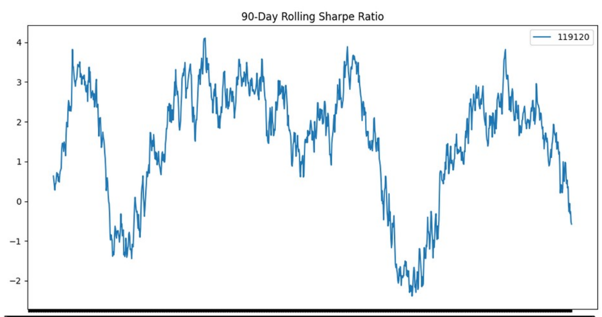

Figure 5.1 — Rolling 90-day Sharpe ratio

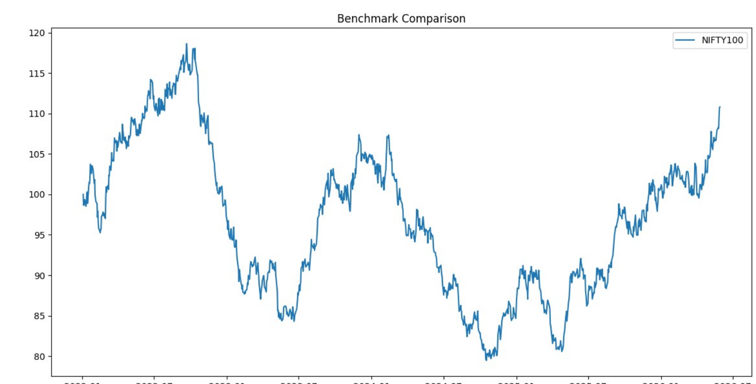

Figure 5.2 — Scheme returns vs. benchmark comparison

5.4 Fund Recommendation Engine

A rule-based recommendation function (Scripts/recommender.py) filters the scorecard by a user-specified risk appetite (matching the risk_grade field) and returns the top-3 schemes by Sharpe ratio within that risk band. This provides a lightweight, explainable starting point for investor-facing recommendations and can be extended with additional filters (e.g., minimum AUM, expense ratio caps, or category preferences) in future iterations.

6. Dashboard

A four-page interactive Power BI dashboard (reports/Bluestock_MF_Dashboard.pbix) was developed to surface the analytics for business stakeholders. The dashboard is organized into Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends pages.

6.1 Industry Overview

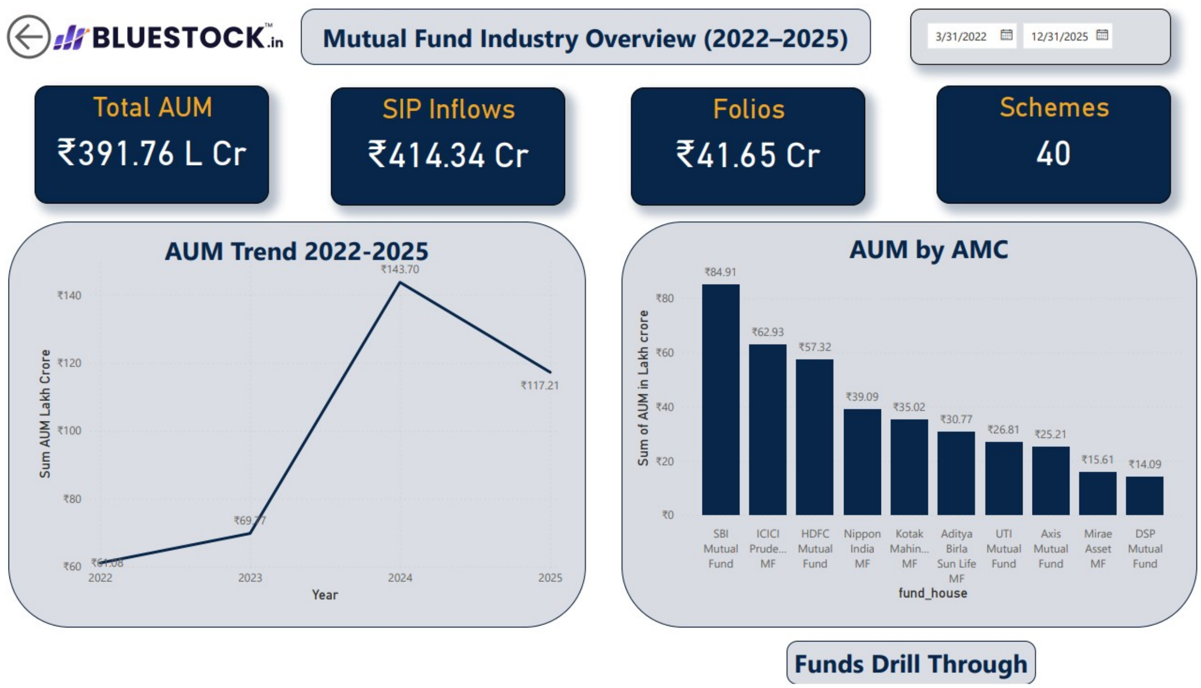

Figure 6.1 — Industry Overview page: AUM trends, SIP inflows, fund-house analysis, and industry KPIs

6.2 Fund Performance

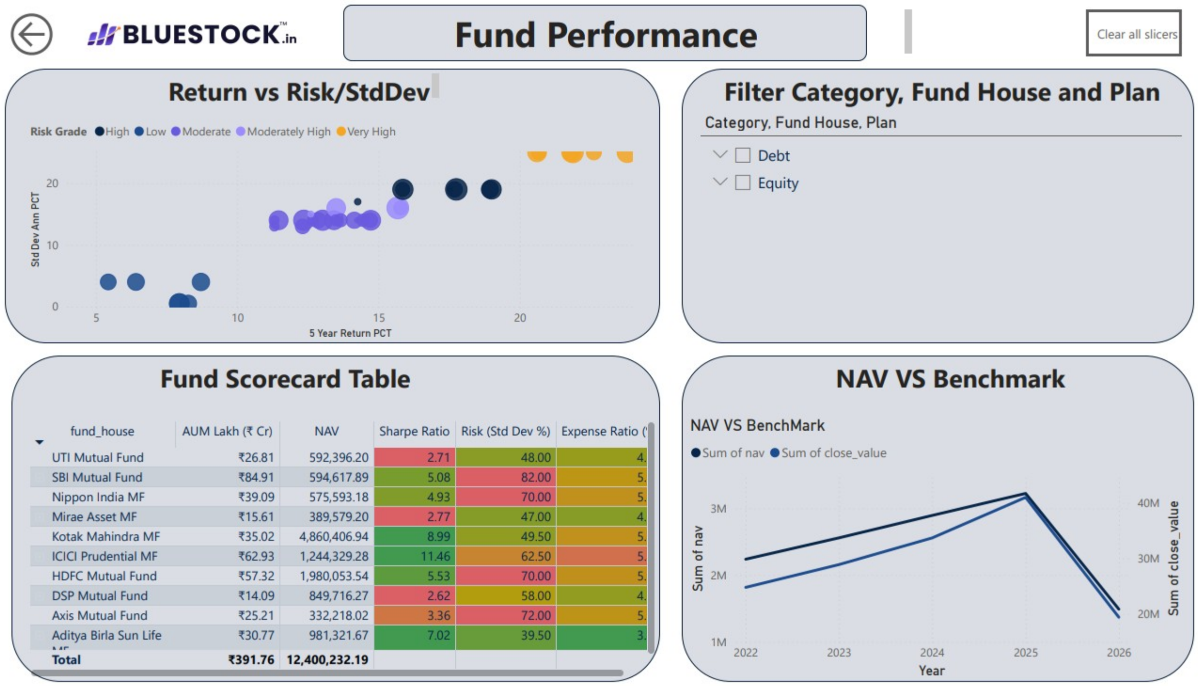

Figure 6.2 — Fund Performance page: risk vs. return, fund scorecard, benchmark comparison, and performance ranking

6.3 Investor Analytics

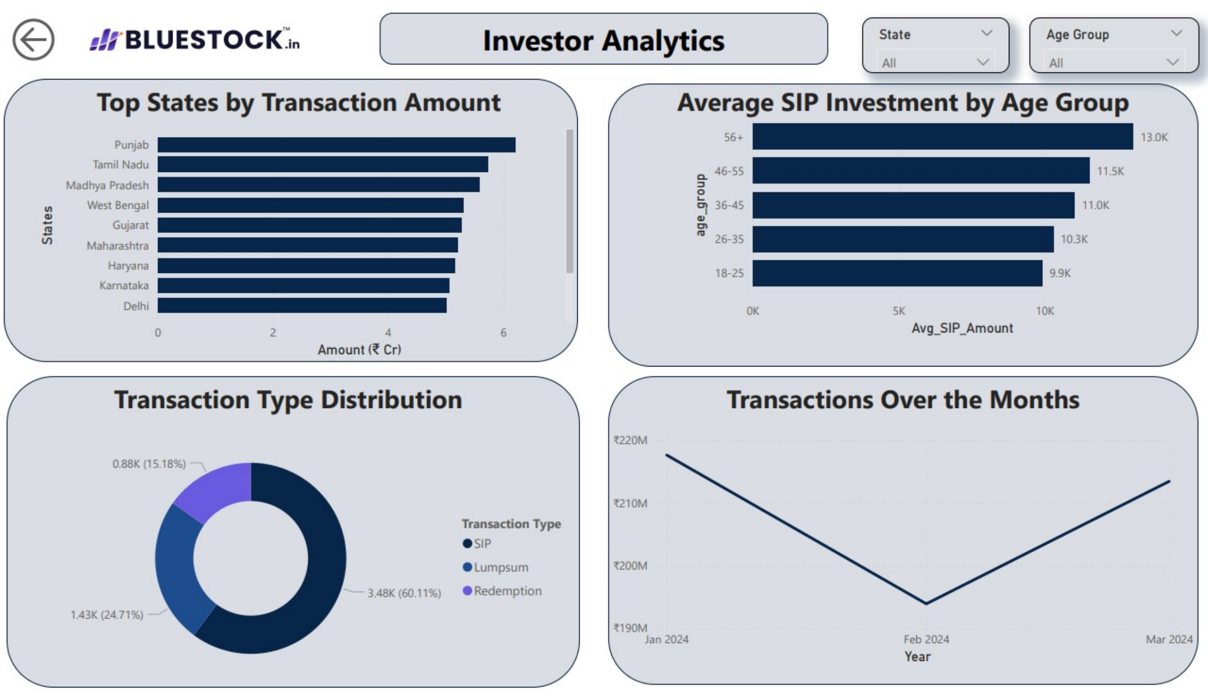

Figure 6.3 — Investor Analytics page: demographics, geographic distribution, and transaction insights

6.4 SIP & Market Trends

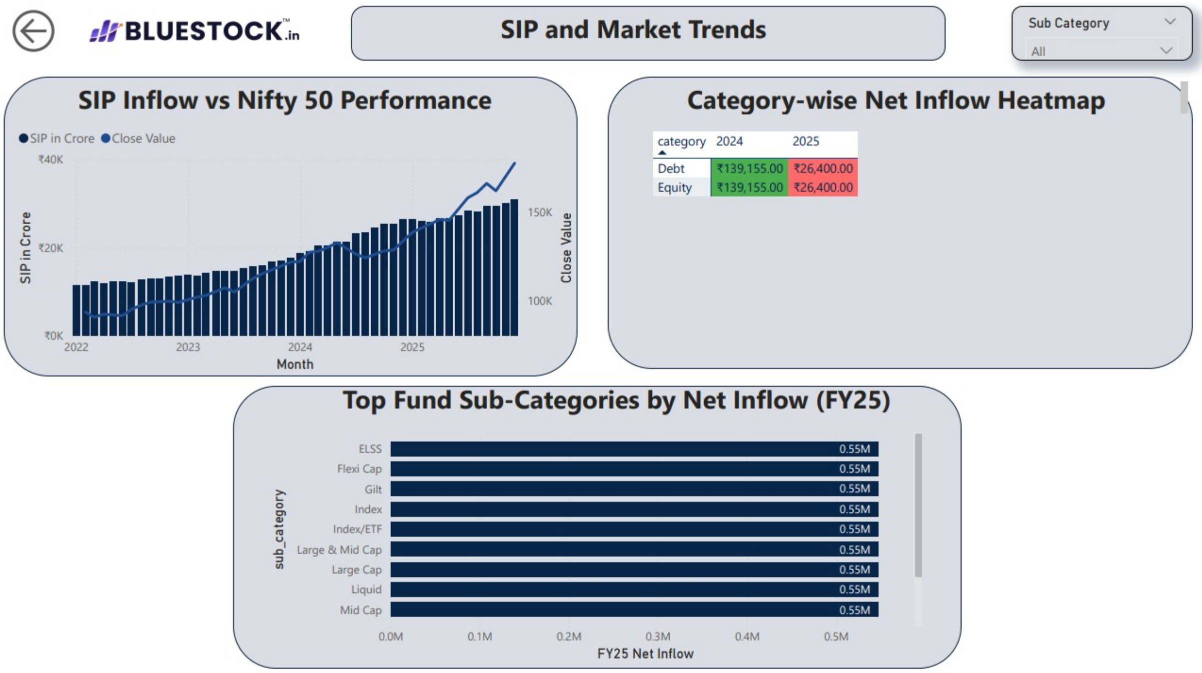

Figure 6.4 — SIP & Market Trends page: SIP inflow vs. Nifty, category inflow heatmaps, and market trend analysis

7. Limitations

- NAV history covering the full 2022-2026 window is available for only six large-cap schemes (fetched live via MFAPI); the remaining 34 schemes in the performance dataset rely on pre-aggregated return and risk metrics rather than full daily NAV series, which limits the depth of return-distribution analysis for those schemes.
- Investor transaction data (08_investor_transactions.csv) is simulated rather than sourced from a live transactional system, so demographic and behavioral insights should be treated as illustrative of plausible patterns rather than verified market facts.
- Performance metrics (CAGR, Sharpe, Sortino, alpha, beta, VaR/CVaR) are point-in-time calculations based on the available historical window and a single benchmark per scheme; results would shift with a different lookback period or benchmark choice.
- The composite Fund Scorecard uses an equal-weighted blend of percentile ranks across five metrics; this weighting scheme is a reasonable default but is not validated against actual investor outcomes or a formal optimization objective.
- The recommendation engine is purely rule-based (risk-grade filter + Sharpe ranking) and does not yet account for an investor's existing portfolio, investment horizon, tax considerations, or liquidity needs.
- The Power BI dashboard is currently distributed as a .pbix file; it has not yet been published to Power BI Service or Tableau Public for web-based access (see Recommendations).
- Sector concentration (HHI) was computed from a single portfolio snapshot date per scheme rather than a time series, so it reflects a point-in-time concentration rather than a trend.

8. Recommendations

8.1 Data & Pipeline Enhancements

- Extend live NAV ingestion via MFAPI to cover all 40 schemes in the performance dataset, enabling consistent daily-return-based risk metrics (VaR, CVaR, rolling Sharpe) across the full universe.
- Replace simulated investor transactions with anonymized real transactional data (or a more sophisticated simulator calibrated to AMFI demographic statistics) to strengthen investor-analytics conclusions.
- Capture portfolio holdings as a time series (monthly snapshots) to enable trend analysis of sector concentration (HHI) and turnover, rather than a single point-in-time view.
- Add automated data-quality assertions (e.g., via a lightweight test suite) to the pipeline so schema or range violations are caught before downstream stages run.

8.2 Analytics Enhancements

- Benchmark Sharpe/Sortino and alpha/beta calculations against category-appropriate indices (e.g., Nifty Smallcap 250 for small-cap schemes) rather than a single broad benchmark, to make alpha/beta figures more directly comparable.
- Introduce a formal weighting/optimization study for the Fund Scorecard — for example, validating the scoring weights against historical forward returns — before using it for investor-facing recommendations.
- Extend the recommendation engine to incorporate investment horizon, expense ratio thresholds, minimum AUM/liquidity filters, and basic portfolio-fit logic (e.g., avoiding over-concentration in a single category or sector).

8.3 Dashboard & Delivery

- Publish the Power BI dashboard to Power BI Service (free account) and add the shareable link to the project README, so stakeholders can access it without opening the .pbix file locally.
- Add drill-through pages for individual schemes (NAV trend, risk metrics, holdings) to support fund-level deep dives directly from the Fund Performance page.
- Schedule periodic data refreshes (e.g., monthly) once live data sources are extended, and document the refresh process in the README for maintainability.

8.4 Documentation & Reproducibility

- Keep run_pipeline.py as the single entry point for end-to-end reproduction, and add basic logging/error handling so partial failures are easy to diagnose.
- Maintain the data dictionary alongside any schema changes so the SQL queries, dashboard, and report remain in sync with the underlying data.

9. Conclusion

This capstone project successfully delivers an end-to-end mutual fund analytics solution spanning data ingestion, cleaning, warehousing, exploratory analysis, performance and risk analytics, and business intelligence reporting. All eight stated objectives — from building a scalable analytics pipeline to developing a recommendation engine and interactive dashboards — have been met using a combination of live API data, industry-style statistics, and a star-schema SQLite warehouse.

The analysis identifies clear, actionable patterns: Small Cap and Flexi Cap categories lead long-term returns but carry higher volatility and drawdown; SIP-based investing continues to grow steadily across both metro and non-metro investors; and a small subset of schemes (led by SBI Small Cap Direct, Kotak Flexicap, and Kotak Emerging Equity) stand out on a composite risk-adjusted basis. These findings, together with the risk (VaR/CVaR, HHI) and recommendation-engine outputs, provide a solid foundation for both investor-facing insights and further analytical development as outlined in the Recommendations section.